Rayleigh Bound State in the Continuum from Hybridized Acoustic Cavities

Author Name[1] and Coauthor Name[1]
[1] *Affiliation to be inserted*
(Dated: August 26, 2026)

A diffraction channel carries no normal flux at a Rayleigh anomaly, but its finite grazing field still prevents localization. We realize a genuine Rayleigh bound state in the continuum in an underwater acoustic lattice. Two dissimilar, individually radiating cavity modes form a Bloch-hybridized band that reaches the first Rayleigh line with vanishing linewidth. At the same wavevector, the pressure amplitudes of both the grazing diffraction order and the coexisting propagating order vanish because the two cavity contributions have equal magnitudes and opposite phases in each channel. Full-band eigenvalue continuation, channel-resolved radiation zeros, and localized eigenfields distinguish this state from a zero-flux Rayleigh anomaly. The result establishes hybridized-cavity double cancellation as a mechanism for Rayleigh BICs in sound.

A bound state in the continuum (BIC) is an eigenstate that remains localized although outgoing waves exist at the same real frequency. Symmetry mismatch, Friedrich–Wintgen interference, Fabry–Pérot trapping, and topological radiation zeros provide established routes to suppressing leakage [1–5]. Their common content is stronger than a large quality factor: every asymptotic component of the *homogeneous* field that fails to decay must vanish.

This distinction becomes singular at a diffraction-channel opening. At a Rayleigh anomaly, the normal wavenumber of one Floquet order is zero, and so is its normal energy flux, but its pressure remains finite and independent of distance from the grating [Fig. 1(b)]. The exterior norm then grows without bound. A Rayleigh anomaly is therefore not a BIC. Electromagnetic reflection gratings have related trapped states to collisions of leaky poles with Rayleigh branch points [6], and recent electromagnetic work has identified Rayleigh BICs through the conjunction of a collective lattice resonance, a channel opening, and a nonradiating response of the constituent scatterer [7]. Scattering matrices at channel openings likewise possess nonanalytic square-root structure [8]. These results establish the threshold as a distinct spectral setting, but, to our knowledge, do not provide the hybridized-cavity mechanism developed here for scalar sound. Here and below, "Rayleigh" denotes a diffraction-order opening rather than an elastic Rayleigh surface wave.

Acoustic BICs have been realized in open resonators, compact scatterers, coupled cavities, and phononic crystals [9–14]. Underwater Fabry–Pérot BICs have also been observed [15]. These advances do not answer the threshold question addressed here: can a *scalar* acoustic eigenmode remain bound exactly when a new diffraction order opens while another order already propagates?

Here we establish such a state using two dissimilar acoustic cavities, neither of which is nonradiating on its own. A broad small-cavity mode and a narrow transverse mode of a large cavity form a Bloch-hybridized band. The band is brought to the $n = -1$ Rayleigh line at an off-Γ wavevector where its radiation vector has a common zero in two independent rows: the finite-flux $n = 0$ order and the grazing $n = -1$ order. The Rayleigh line therefore fixes the spectral endpoint, whereas cavity hybridization supplies the amplitude and phase freedom needed to eliminate both nondecaying fields. The resulting state is a Rayleigh BIC not because the threshold flux vanishes, but because the threshold field itself does.

Localization at a channel opening.— Consider a rigid surface periodic along x, with water in $y > 0$ and time dependence $e^{+i\omega t}$. A homogeneous outgoing field is

$$p^{\mathrm{hom}}(x,y) = \sum_{n\in\mathbb{Z}} A_n e^{-ik_{x,n}x - ik_{y,n}y}, \quad k_{x,n} = k_x + \frac{2\pi n}{a}, \tag{1}$$

where $k_{y,n} = \sqrt{k_0^2 - k_{x,n}^2}$ is chosen outgoing or decaying. Introducing $\Omega = af/c$ and $\kappa = k_x a/(2\pi)$, the nth Rayleigh line is $|\kappa + n| = \Omega$. For one exterior order,

$$\int_0^H |p_n|^2\,dy = \begin{cases} H|A_n|^2, & k_{y,n} = 0, \\ \dfrac{|A_n|^2}{2\alpha_n}(1 - e^{-2\alpha_n H}), & k_{y,n} = -i\alpha_n. \end{cases} \tag{2}$$

Equation (2) is the decisive threshold criterion: zero normal flux is insufficient; square integrability requires $A_n = 0$ for a grazing order, just as it requires zero amplitude for every propagating order.

Acoustic realization.— The unit cell [Fig. 2(a)] contains two water-filled grooves in a rigid reflector. At $c = 1500\ \mathrm{m\,s^{-1}}$, the dimensional realization has $a = 7.4167$ mm and a Rayleigh BIC at

$$\begin{gathered} f_{\mathrm{BIC}} = 180.000\ \text{kHz}, \\ (\kappa_{\mathrm{BIC}}, \Omega_{\mathrm{BIC}}) = (0.1100001053,\ 0.8899998947), \end{gathered} \tag{3}$$

corresponding to $\theta_{\mathrm{BIC}} = 7.09966°$. At positive κ, $n = 0$ is propagating, $n = -1$ is grazing, and $n = +1$ is evanescent. The reciprocal partner at negative κ exchanges the two first orders.

We solve a pole-free modal-matching eigenproblem retaining the exterior Floquet amplitudes $\boldsymbol{A}$ and all trans-

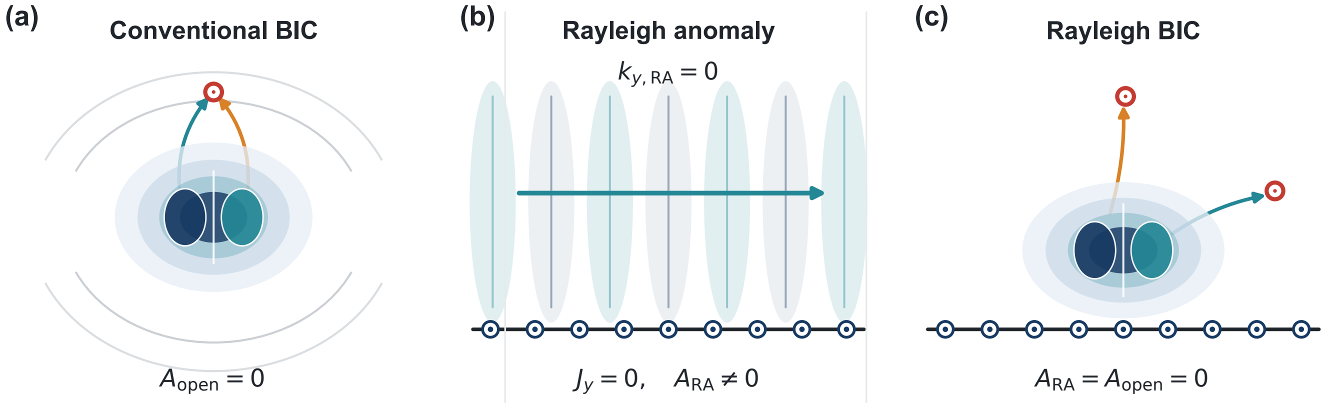

FIG. 1. From a conventional BIC to a Rayleigh BIC. (a) A conventional BIC cancels all open radiation. (b) At a Rayleigh anomaly the grazing order has $k_{y,\mathrm{RA}} = 0$ and zero normal flux, but a finite nondecaying amplitude A_{RA}; the state is not localized. (c) A Rayleigh BIC requires simultaneous cancellation of the grazing amplitude and every remaining open amplitude.

verse groove coefficients $\boldsymbol{C}$,

$$\mathsf{F}(\Omega,\kappa)\begin{bmatrix}\boldsymbol{A}\\ \boldsymbol{C}\end{bmatrix} = 0. \tag{4}$$

The full dispersion in Fig. 2(b) shows the target branch connecting a symmetry-protected BIC at Γ to the off-Γ Rayleigh BIC at Eq. (3). The latter occurs when the branch reaches the $n = -1$ Rayleigh line; its imaginary frequency vanishes and the quality factor diverges [Fig. 2(c)]. With 345 exterior orders and 43 transverse modes per groove, the full operator at the exact endpoint has $\sigma_{\min}/\sigma_{\max} = 5.85 \times 10^{-16}$ and raw residual 1.28×10^{-13}. More importantly, its outgoing eigenvector satisfies

$$A_0 = A_{-1} = 0, \tag{5}$$

while all retained exterior components are evanescent. Channel-resolved fields at the Γ BIC, the Rayleigh BIC, and beyond the opening show the corresponding channel topology; at the Rayleigh crossing, neither nondecaying order survives [Fig. 2(d,e)]. Thus the diverging Q, the real-axis eigenvalue, and the localized eigenfield are three views of the same homogeneous state.

Origin of the double zero.— To expose the mechanism without replacing the full eigenproblem, we resolve its aperture field into two dressed coordinates, $\boldsymbol{u} = (u_L, u_S)^T$, dominated by the large- and small-groove mode families. The groove-resolved boundary sources in the two relevant orders can be written

$$\begin{bmatrix}J_0\\ J_{-1}\end{bmatrix} = \mathsf{D}_s(\kappa)\boldsymbol{u}, \qquad \mathsf{D}_s = \begin{bmatrix}d_{0L} & d_{0S}\\ d_{-1,L} & d_{-1,S}\end{bmatrix}. \tag{6}$$

For the propagating order, velocity continuity gives $A_0 = -(k_0/k_{y,0})J_0$; hence $J_0 = 0$ closes that channel. At the Rayleigh threshold, $J_{-1} = 0$ enforces zero normal velocity but does not by itself imply $A_{-1} = 0$. The latter pressure condition is supplied independently by the complete eigenvector in Eq. (5). The source reduction therefore explains the cancellation mechanism, while the full operator establishes strict localization.

The Bloch phase changes both hybrid weights and their relative phase [Fig. 3(a,b)]. At κ_{BIC}, the two groove contributions $j_{mL} = d_{mL}u_L$ and $j_{mS} = d_{mS}u_S$ obey

$$|j_{mL}| = |j_{mS}|,\qquad \arg\left(\frac{j_{mL}}{j_{mS}}\right) = \pi, \qquad m = 0, -1. \tag{7}$$

Thus $J_0 = J_{-1} = 0$ at the same Bloch wavevector [Fig. 3(c,d)]. Neither isolated groove is nonradiating; periodicity instead selects a hybrid cavity eigenvector whose channel sources cancel in both rows, while the exact pressure amplitudes simultaneously obey Eq. (5). Calculations for the isolated grooves, the nonperiodic dimer, and a bounded single-groove search are given in the Supplemental Material.

Scattering observables.— A strict BIC is dark under an ideal plane-wave drive; it is inferred from the collapse of the neighboring quasi-BIC response rather than from a finite field exactly at the BIC. Figure 4(b) therefore plots the specular phase in coordinates centered on the complex pole. The phase feature narrows to zero at θ_{BIC}, while the resonant parts of the $n = 0$ and $n = -1$ scattering amplitudes close simultaneously [Fig. 4(c,d)]. A complex pressure scan above the sample can be Fourier projected onto the Floquet orders, providing a direct channel-resolved measurement [Fig. 4(a,e)].

In the present theory draft, the open symbols and the coarsely sampled field are exact subsamples of the calculations, chosen on the intended experimental grid. They define the data products that must be replaced by measurements; they are not presented as experimental validation.

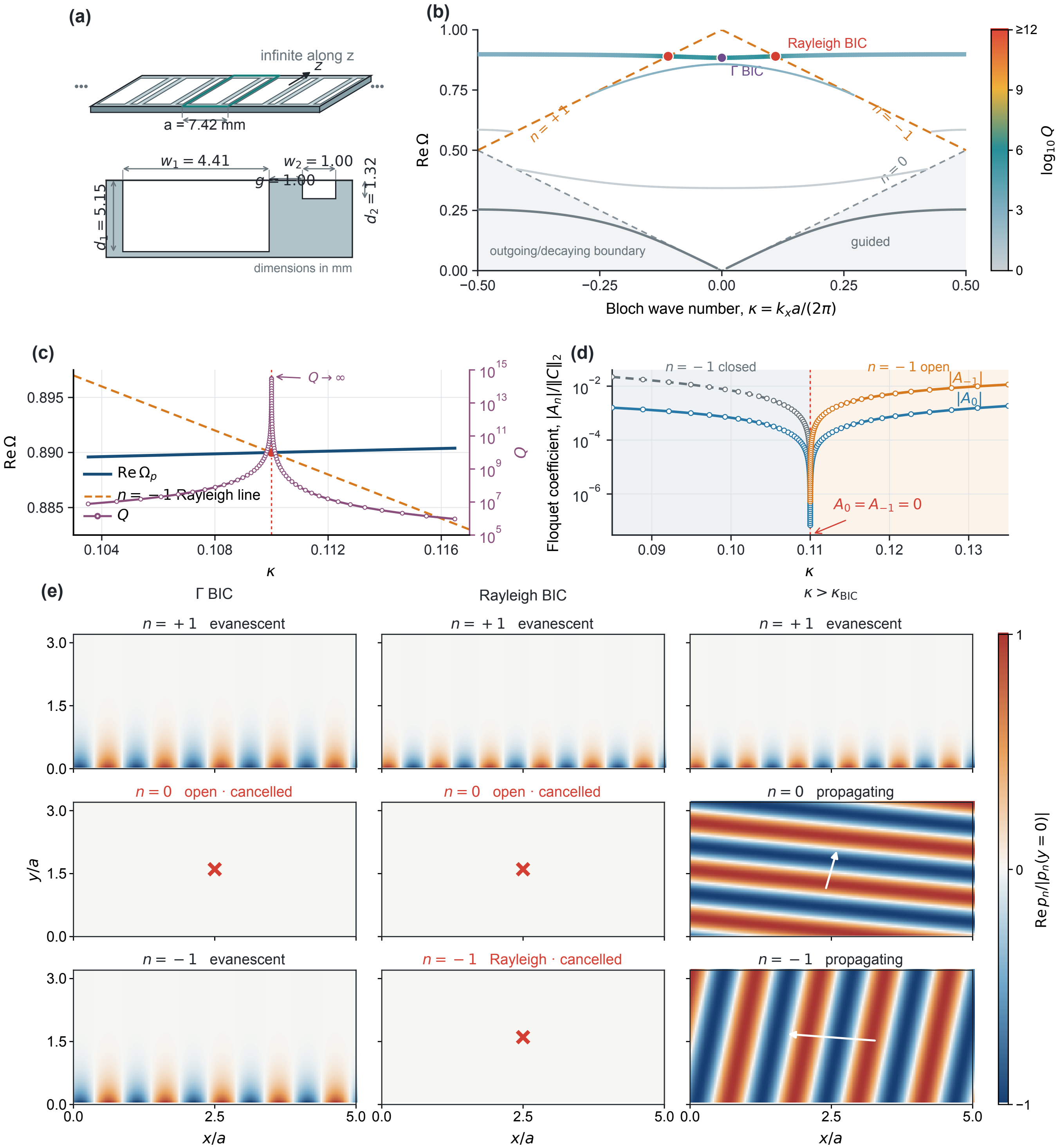

FIG. 2. Rayleigh BIC in an underwater acoustic metagrating. (a) Periodic two-groove surface and unit-cell dimensions. (b) Complex acoustic band structure; color gives $\log_{10} Q$, with vivid colors denoting high Q. The purple point is a symmetry-protected Γ BIC, whereas the red points mark the reciprocal Rayleigh-BIC pair. (c) Local branch: $\mathrm{Re}\,\Omega_p$ reaches the $n = -1$ Rayleigh line as Q diverges. (d) Homogeneous Floquet pressure amplitudes: A_0 and A_{-1} vanish together. (e) Order-resolved exterior eigenfields at the Γ BIC, the Rayleigh BIC, and beyond the opening.

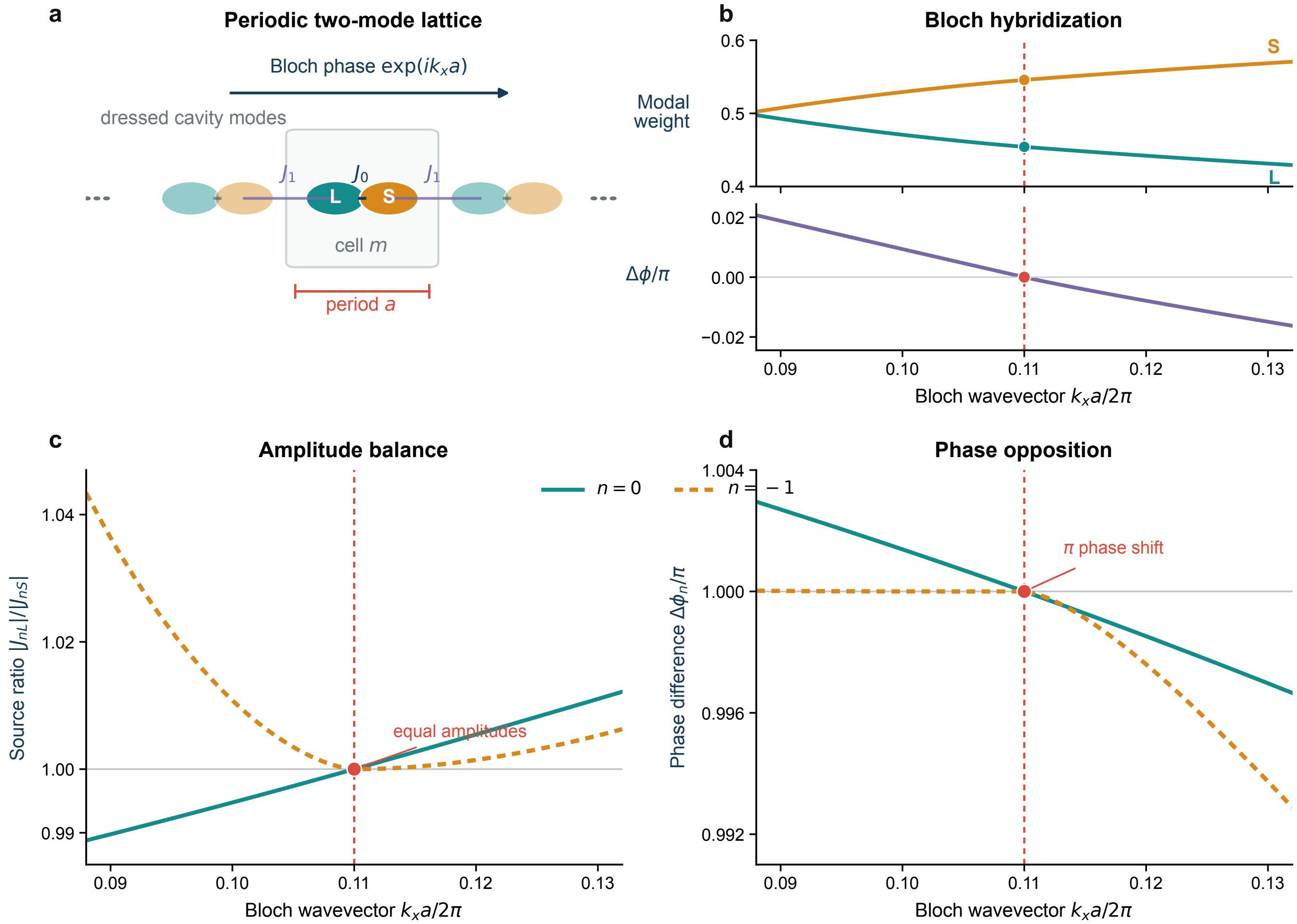

FIG. 3. Two-coordinate source reduction of the Rayleigh BIC. (a) Periodic cavity lattice with intracell and intercell coupling. (b) Bloch hybridization changes the large- and small-cavity weights and their relative phase. (c,d) At the BIC, the two groove source terms have equal magnitudes and a π phase difference in both the $n = 0$ and $n = -1$ radiation rows. The strict pressure zeros are obtained from the full eigenproblem in Fig. 2.

The result identifies a mechanism for a Rayleigh BIC in an underwater acoustic lattice. The vanishing channel velocity explains the zero normal flux, but not localization. Localization emerges only when a hybridized cavity Bloch eigenvector is simultaneously orthogonal to the grazing and propagating pressure-radiation rows. The two-groove surface realizes this common null through interference between a narrow transverse cavity mode and a broad radiative cavity mode. Consequently, the central state is neither a generic Rayleigh anomaly nor a merely high-Q resonance: it is a Rayleigh BIC in a sound-wave continuum, created when a hybridized cavity band reaches the diffraction threshold with both nondecaying radiation amplitudes canceled.

The ideal statement applies to an infinite, rigid, lossless grating. Finite aperture, material compliance, viscous attenuation, and disorder turn the BIC into a quasi-BIC and bound the experimentally accessible Q. These effects do not alter the amplitude-level criterion in Eq. (5), but they must be included in a quantitative comparison with a finite water-tank sample.

Acknowledgments and funding information to be inserted.

[1] H. Friedrich and D. Wintgen, Physical realization of bound states in the continuum, Physical Review A **31**, 3964 (1985).
[2] C. W. Hsu, B. Zhen, A. D. Stone, J. D. Joannopoulos, and M. Soljačić, Bound states in the continuum, Nature Reviews Materials **1**, 16048 (2016).
[3] C. W. Hsu, B. Zhen, J. Lee, S.-L. Chua, S. G. Johnson, J. D. Joannopoulos, and M. Soljačić, Observation of trapped light within the radiation continuum, Nature **499**, 188 (2013).
[4] Y. Yang, C. Peng, Y. Liang, Z. Li, and S. Noda, Analytical perspective for bound states in the continuum

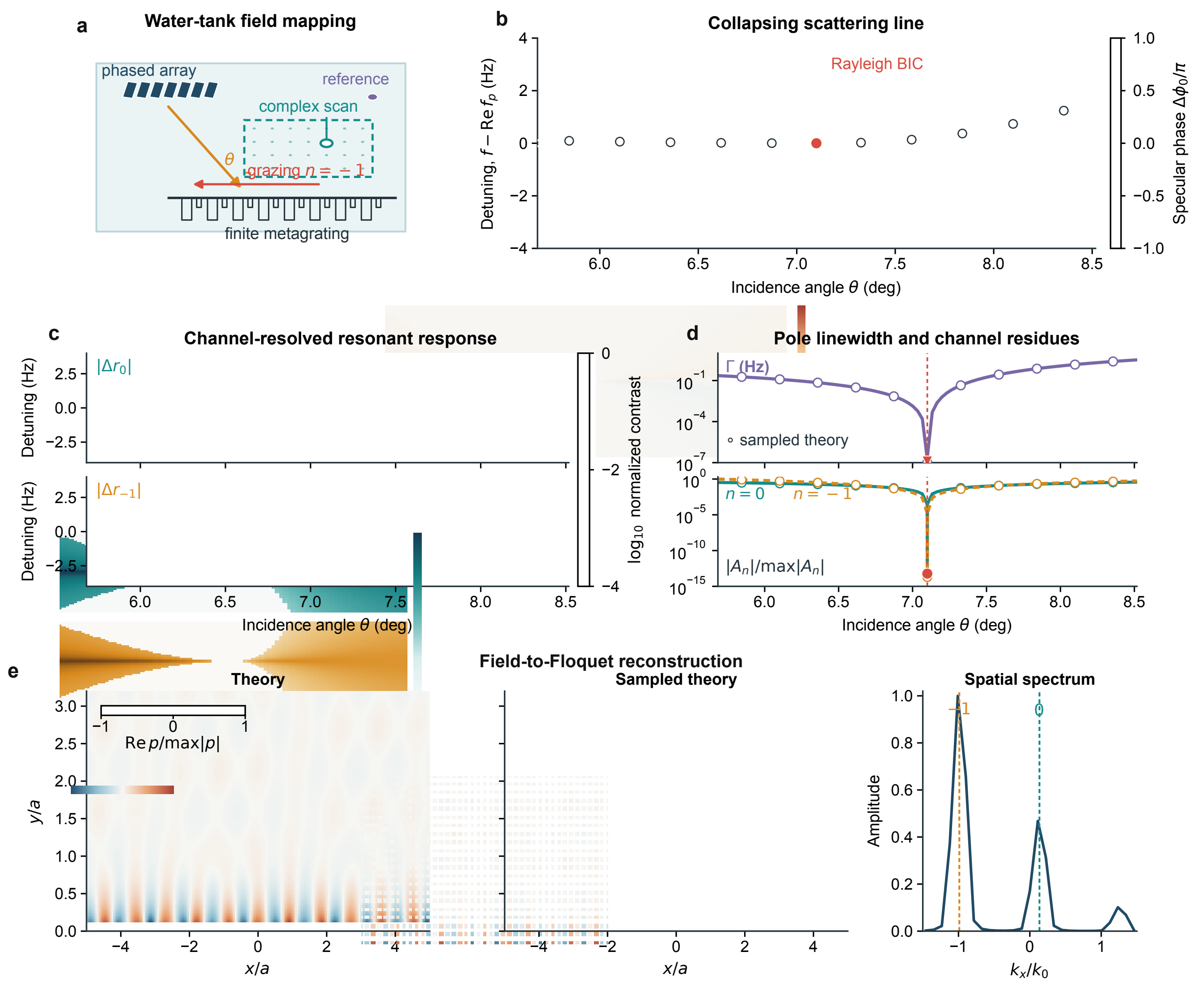

FIG. 4. Observable scattering signatures. (a) Water-tank complex-field mapping. (b) Collapse of the resonant specular-phase line. (c) Channel-resolved resonant contrasts for $n = 0$ and $n = -1$. (d) Pole linewidth and homogeneous channel residues. (e) Complex-field sampling and Floquet reconstruction at a nearby driven state; an exact BIC cannot be excited by the external plane wave. Open symbols and the sampled field are theory subsamples reserved for future measured data.

in photonic crystal slabs, Physical Review Letters **113**, 037401 (2014).

[5] B. Zhen, C. W. Hsu, L. Lu, A. D. Stone, and M. Soljačić, Topological nature of optical bound states in the continuum, Physical Review Letters **113**, 257401 (2014).

[6] F. Monticone and A. Alù, Bound states within the radiation continuum in diffraction gratings and the role of leaky modes, New Journal of Physics **19**, 093011 (2017).

[7] I. Karavaev, M. Song, and A. Bogdanov, Rayleigh bound states in the continuum (2026), arXiv:2607.07228v1, arXiv:2607.07228 [physics.optics].

[8] C. C. Wojcik, H. Wang, M. Orenstein, and S. Fan, Universal behavior of the scattering matrix near thresholds in photonics, Physical Review Letters **127**, 277401 (2021).

[9] L. Huang, Y. K. Chiang, S. Huang, C. Shen, F. Deng, Y. Cheng, B. Jia, Y. Li, D. A. Powell, and A. E. Miroshnichenko, Sound trapping in an open resonator, Nature Communications **12**, 4819 (2021).

[10] L. Huang, B. Jia, A. S. Pilipchuk, Y. Chiang, S. Huang, J. Li, C. Shen, E. N. Bulgakov, F. Deng, D. A. Powell, S. A. Cummer, Y. Li, A. F. Sadreev, and A. E. Miroshnichenko, General framework of bound states in the continuum in an open acoustic resonator, Physical Review Applied **18**, 054021 (2022).

[11] I. Deriy, I. Toftul, M. Petrov, and A. Bogdanov, Bound states in the continuum in compact acoustic resonators, Physical Review Letters **128**, 084301 (2022).

[12] M. Martí-Sabaté, J. Li, B. Djafari-Rouhani, S. A. Cummer, and D. Torrent, Observation of two-dimensional acoustic bound states in the continuum, Communications Physics **7**, 122 (2024).

[13] M. Krasikova, F. Kronowetter, S. Krasikov, M. Kuzmin,

M. Maeder, T. Yang, A. Melnikov, S. Marburg, and A. Bogdanov, Acoustic bound states in the continuum in coupled Helmholtz resonators, Physical Review Applied **22**, 024045 (2024).

[14] L. Yang, R. Zheng, S. Zhang, W. Zhang, Q. Du, P. Peng, Z. Wang, M. Ke, X. Huang, and F. Liu, Lines of bound states in the continuum in a phononic crystal slab, Physical Review B **111**, 134102 (2025).

[15] Y. Yang, H. Jia, Y. Yang, P. Zhou, and J. Yang, Observation of merged Fabry–Perot bound states in the continuum within an underwater phononic crystal, Physical Review B **110**, 184108 (2024).